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Experimentalism and invention

CONSIDER LOUIS PASTEUR'S swan-necked flasks, deployed in public experiments in mid-nineteenth-century Paris. The drawn-out necks were bent so that any air passing into the flasks would deposit its 'dust' or organic particles on the bend of the neck; it would not reach the broth in the bowl of the flask. The design was a clever solution to a problem in fermentation experiments – how might you allow the access of air without the access of dust? By this means Pasteur showed that 'clean' air did not cause fermentation but dust did. By many such experiments he satisfied his audiences that he could 'control' fermentation. It was *not* a spontaneous process; microbes were not spontaneously generated (at least under everyday conditions); microbes entered from the air and they caused fermentation in organic broths. Of course, you could not prove that microbe-generation was never, ever spontaneous, but you could marginalise questions about the ultimate origin of microbes (or diseases); you could move them beyond the normal concerns of chemists, brewers, hygienists and doctors.¹

This control over fermentation served to explain some inventions and to provide many new ones. It made clear why, if you boiled broths and jellies in full vessels and then capped them to exclude air, the fluids would usually 'keep' until the jar was opened. This was the principle of 'canning', which proved so important for the importation of meat from South America. And the heating routines which Pasteur worked out were used to sterilise, or pasteurise, milk or wine with minimal change to the taste.

Much of Pasteur's work was rooted in industrial problems; his work on fermentation derived in part from alliances with beer-makers in Lille, and his later work on diseases from epidemics which killed silkworms. Arguably, the force of Pasteur's research programmes derived from his training in chemical experimentalism which he applied to 'model diseases' of wine and beer. Fermentations, he argued, were biological rather than chemical, but the techniques were not far removed from his chemical studies. Then he pursued his experimentalism into diseases of animals, when human diseases, for obvious reasons, were not the subjects of direct experiment. Pasteur exploited 'biotechnologies' which could be manipulated – his 'models' gave rise to practices as well as principles. That work continued in the Pasteur Institutes erected to his honour in Paris and the French colonies; their studies of bacteria and vaccines were 'state of the art' in biomedicine around 1900.

This chapter argues that experimental science was largely a product of the nineteenth century. It stresses the relation between experiment and systematised invention; it concentrates on the creation and control of novelty. My technique is to build on historical writings that are regarded as sound and exciting, yet which remain peripheral to our received accounts of science. Here again, I extend some of Kuhn's (essential tension) arguments, especially by including the biomedical sciences as running in parallel with physical sciences. Again, I stress that I am less concerned with origins than with innovations, i.e. the entry of inventions into social practice, their becoming part of the workings of the world. So we can afford Greek ancestors and early modern progenitors, without locating there the social innovations which belong to times and places much closer to our own.

At the end of the chapter I consider the extent to which my account of experimentalism can also be applied to invention, as a way of making which developed contemporaneously, as part of the division of labour in nineteenth-century industry. I shall argue that the parallels between experiment and invention were so close that the two might be regarded as the sides of one coin.

This finding will then provide the basis of the following chapter – on technoscience.

Meanings of experiment

Though a few philosophers of science have focused on experimentation (Hacking, 1983), and several historians and sociologists of science have written at length on particular experiments (Collins, 1992; Galison, 1987; 1997; Gooding, Pinch and Schaffer, 1989; Shapin and Schaffer, 1985), we lack a commonly agreed typology of experiments. (This is another reflection of the monism of most philosophy of science and its sociological stepchildren – the assumption that science is much the same, anytime, anyplace – or that the differences are so many and various that nothing systematic can be said about them.) But we can draw on Kuhn (1977b) and the distinctions made by early methodologists who were concerned with promoting experiment.

Many early modern ‘experiments’ in the classical sciences were essentially *demonstrations*. If classical sciences, especially mechanics, can be characterised as analytical in reducing complex motions to a few simple principles, then this role of experimentation is easily understandable. The demonstrator creates a simple situation in which the elementary motions can be observed free of their usual (natural) complications. It is also evident why there may be a tendency to conflate real experiments with ‘thought experiments’, conducted ‘in the head’ or ‘on paper’. Such experiments, like classroom experiments, are not meant to produce anything new, nor even to allow decisions between hypotheses – they are exercises in working out analytical principles to reconstruct phenomena, in principle, from their elements. Though experimentation, even in this sense, was not a significant part of ancient planetary astronomy, it played a role in optics, hydrostatics and statics. It was important in medieval studies of local terrestrial motion as an area of quantitative analysis. Historians dispute the extent to which the experiments

of Galileo are to be read as demonstrative rather than as tests of hypotheses.

Experimental histories

Kuhn was keen to distinguish this demonstrative role of experiment from the tendency he saw developing in the sixteenth and seventeenth centuries, described and encouraged by Francis Bacon. Nature was to be 'tortured', preferably by instruments, so as to give up her secrets – to produce effects not to be seen in the normal course of events. He links this tendency to the fading of the distinction between nature and artifice (as discussed in Chapter 3). If God was a craftsman, then his craft work could be taken apart or put to the test in ways that were continuous with the exertions of human craftsmen. Bacon distinguished between experiments meant to be of direct economic benefit and 'experiments of light' that were meant to illuminate the phenomena; but we can follow Kuhn in seeing most such experimentation as largely separate from experiment/demonstration in the analytical sciences (Shapin, 1994; Shapin and Schaffer, 1985). Baconian experiments were meant to feed into crafts, or to provide 'experimental histories' which were part of natural history (in the wide sense). Indeed, we may see Baconian experimentation within that early modern reformulation of natural history which 'cast out' the hermeneutic and 'cast in' the 'history of nature-worked-on-by-man'. These enlargements of the field of natural history were also related to the expansion of range made possible by telescopes, microscopes and voyages of exploration. However, the field so recast and expanded remained natural history in as much as the phenomena could only be described and catalogued.

In these respects, my classification seems in line with the detailed analyses of early modern experimentation published by Shapin and Schaffer to which I have briefly referred in earlier chapters. They make the connection between the air pump and the new optical instruments as ways of making phenomena visible; they show how Robert Boyle avoided mathematics and

linked his experiments to natural history; above all, they stress the production of 'facts' by public witnessing and reporting. As several scholars have argued, the status of 'facts' was also crucial for the new natural history (e.g. Daston and Park, 1998). There is then ample mandate for seeing the air pump as a contribution to 'experimental history'.

But, because I do not see such 'experiments' and natural history as the only activities characteristic of science in that period, there is also room for other readings. The dynamics of mathematics, or more generally of the analytical sciences such as planetary astronomy and mechanics, seem obviously different from those of natural history, as I have briefly explored in Chapter 4. To the extent that Boyle's studies of air involved the testing of hypotheses about the 'mechanics' of air – its 'spring' and pressure-to-volume ratios, they might be said to be grounded in an analytical reduction of 'air', and to be comparable to analytical experimentation in mechanics. Such experiments could be demonstrative, and/or measurements to establish relationships – like Galileo rolling balls down inclined planes as a relatively easy way of 'timing' falling bodies.

So perhaps it would be useful to see the air pump experiments in *several* perspectives – as natural-historical *and* analytical, and maybe also with some characteristics of the 'synthetic experimentation' on which this chapter concentrates. My suggestion is that most early modern experimentalism was subordinate to natural history, or to the limited analytical sciences. My guess is that experimentalism could only become a major way of knowing when 'synthesis of elements' became a way of creating and controlling novelties. In that sense, nineteenth-century experimentalism was built on the successes of analysis from the end of the eighteenth.

Experimentation and the age of analysis

In discussing analysis in the decades around 1800, we have seen the importance of quantification, especially in the formation of

new chemical and physical disciplines. Weight was central to Lavoisier's chemistry, including the establishment of the combining-weights for each of the elements. Such measurements required considerable technical skill and equipment – both the torture instruments of the chemist's art (furnaces, condensers, etc.) and the instruments of measurement, such as balances, which had been more associated with mathematical demonstrations or perhaps with the commercial side of chemistry. Similarly, mathematicians, philosophers and engineers who analysed such physical phenomena as the flow of heat, used sophisticated thermometers and elaborate ways of protecting the 'model' from variations in external temperature (and the body heat of the experimenter, etc.). Reconstructions of J. P. Joule's paddle-wheel experiments have shown the difficulty of the task, and the way Joule drew on his 'domestic' advantages – his familiarity with thermometers used in brewing (as methods were standardised), and a cellar where the temperature was more or less constant throughout the day (Sibum, 1995). The themes of accurate measurement and protected conditions have also been well explored for academic physics laboratories in the nineteenth century. Graeme Gooday has shown how teachers of physics in the then new universities of industrial England (and in Cambridge and Glasgow) stressed accurate measurement as a discipline on which students could draw in their later (industrial) lives (Gooday, 1990; 1991a; 1995).

Of the biomedical sciences *c.*1900, physiology was the discipline most associated with experimentation, and historians of that discipline have often drawn a fairly direct line back from the twentieth century to Claude Bernard (*c.*1860), to François Magendie (*c.*1820), to Xavier Bichat (*c.*1800) and to Albrecht von Haller (*c.*1760). But here too, much of the experimentation or practical work might better be seen as forms of analysis and measurement. Haller attempted dynamic characterisation of body parts in terms of their 'sensibility' and 'contractility'. With Bichat, as noted in the previous chapter, this kind of dynamic classification was extended over a range of tissues/systems explicitly presented as the elements of the body. This 'general

physiology' continued to be a major part of physiology in the nineteenth century, including analyses of reactivity (irritability, etc.) as well as histological and chemical analyses of body tissues (Lesch, 1984). Likewise, much of the 'physical physiology', in the new laboratories of German universities c.1840 could be seen as continuing this tradition of 'characterisation of elements', e.g. electrical studies of muscle and nerve. We easily forget that the first physiology institutes were much concerned with analytical microscopy or physiological chemistry; indeed, the wish of medical students to acquire facility in these techniques of medical analysis was one of the selling points of the new physiology (Kremer, 1992; Tuchman, 1993). Such techniques may be seen as analogous (and continuous) with the analytic and measurement skills promoted by chemical and physical institutions.

I do not claim that all early nineteenth-century physiology was analytical, but it was chiefly after mid-century, especially through Claude Bernard's writings in the 1860s, that experimentalism as a method was clearly formulated. Curiously (and this fact seems to have escaped previous commentators), Bernard's innovations closely followed the corresponding reflections of his friend and colleague at the Collège de France, the chemist Marcelin Berthelot (Crosland, 1970). To illuminate this nexus, and the much wider questions of self-conscious experimentalism, we turn to consider the methodologies of nineteenth-century chemistry, then of biological sciences and then of 'physics'.

Synthesis in chemistry

I have already suggested that synthesis in chemistry was the analogue of experimentalism in biomedical sciences. For me, initially, this analogy was a hypothesis: I was looking for analogues of Bernardian experimentalism; I found my hypothesis confirmed in interesting ways for French science, and then I learned something of the larger history of 'synthesis' in chemistry. Though historians of chemistry had written about synthesis, it

was usually as a prelude to the fine chemical industry, or in relation to the once vexed question of how the characteristic molecules of *living* bodies were related to the simpler substances of which they were composed. But synthesis was also important as a scientific method, a way of knowing. As Berthelot repeatedly argued, synthesis took chemistry beyond mere analysis; it afforded a means of checking analysis, but also of testing predictive hypotheses; and it also allowed the creation of existing useful chemicals, or of chemicals unknown in nature (Berthelot, 1879). Berthelot was a major proponent of chemistry applied to industry.

However, the vogue for synthesis predated the discussions in France. It was in industrial Britain that chemistry was first promoted for its synthetic possibilities – but the Britons were led by a German. Wilhelm Hofmann, Liebig's favourite disciple, was the director of the Royal College of Chemistry, founded in London in 1845 and state funded from 1853. He promoted this pioneering institution not just by training analysts, but by pointing to the possibilities of synthesising new materials. Interestingly, atomic models were central to his demonstrations; wooden balls joined by sticks allowed chemists to envisage how atoms might be arranged in compounds. If you could model natural compounds, you might extend your models to new compounds and then try to make them. Models encouraged ideas of 'building' new molecules. It was all very practical, concrete and British: the models were based on the atomic theory pioneered by John Dalton at the opening of the century. Amusingly, when Hofmann returned to Germany as a university professor, he dropped modelling as unsuited to the 'higher', more philosophical approaches to science (Meinel, forthcoming).

Berthelot, by contrast, had never been fond of models. Philosophically, like Claude Bernard, he was a follower of Auguste Comte, the founder of positivism. For positivists, science was the discovery of laws (not crude models of structures that could not be known directly). But positivism, too, was easily compatible with an enthusiasm for synthesis as a way of confirming hypotheses; indeed, for many positivists, the ability to

manipulate nature predictably was good evidence of the correctness of their scientific principles. As we shall see in the case of the biologist, Jacques Loeb, a positivistic, 'engineering' approach to science could mean a heavy stress on experiment as the highest mode of investigating life (Pauly, 1987).

It seems significant that both Berthelot and Bernard worked at the Collège de France, a privileged Parisian facility for research and special lecture courses, and *not* for the training of practitioners; both argued for the utility of their disciplines by demonstrations of experimental synthesis and control, rather than by analysing economic or medical processes. And the other famous professor at the college was none other than Louis Pasteur, who in the 1860s became widely known for his research on the causes of fermentation; his control experiments using swan-necked flasks, with which I introduced this chapter, were to become classics of 'scientific method'. But Berthelot had yet another reason for stressing 'synthesis'. He wrote a history of chemistry which featured Lavoisier as the proponent of analysis, thus making dialectical space for Berthelot himself to appear as the chief proponent of chemistry's next stage (Crosland, 1970).

Experimentation in biomedical sciences

What Berthelot claimed for synthesis in chemistry, Claude Bernard claimed for controlled experiments in physiology: they were the means of moving biological sciences beyond the merely analytical to a higher level of scientific method, the means of testing predictive hypotheses. Most commentators have stressed the obvious: that Bernard was pleading the cause of his own discipline, physiology, which he presented as 'experimental medicine'. They have noted his denigration of existing medical sciences as being '*merely descriptive*', including statistical studies in medicine – the 'numerical method' mentioned in my previous chapter. Bernard's criticism often surprises present-day readers, because statistics are now accepted as an essential aspect of experimentalism, but he was convinced that one well-controlled

experiment could decide issues in a way that transcended statistical observations (Canguilhem, 1975; Grmek, 1970; Holmes, 1974).

Commentators have also noted Bernard's attachment to the positivism of Auguste Comte, including the characteristic advocacy of biology as a general science of life, and some have stressed the multiple meanings of 'control' in Bernard's work (Figlio, 1977). Controlled experiment in his new, technical sense, involved 'controls' – systems that were just like the real experiment except for the key factor under investigation. But this was also a way of 'controlling' the experiment in a more general sense, a means of gaining control over living processes. The promise of Bernard's *Experimental Medicine* (see Bernard, 1957) was clear and persuasive: what could be controlled in laboratory animals today might be controlled in human patients tomorrow. Animal experiments would lead to a dynamic understanding of living processes, and on to therapies. And so it proved – after about fifty years of work.

There is room for more investigation of the Parisian context of these arguments. We know that some zoologists objected to Bernard's 'put down' of their science; one of them argued that he could test hypotheses by searching out 'natural experiments' where the operation of the key factor could be observed (Churchill, 1973). In an earlier debate, M. E. Chevreul, the Natural History Museum chemist, famous for his work on colours and on painting, had pointed out the role of '*a posteriori*' experiments in testing hypotheses. For example, if you wanted to know why dyes had different effects depending on the composition of the water, you could analyse the waters and guess the key factors; but then you should test the hypothesis by adding the critical factor to distilled water to see if you got the expected result. The effects of spa waters on human ailments could, in principle, be explored by similar methods (Churchill, 1973; and see Chevreul, 1824). That such matters were discussed in the Paris Academy of Sciences may suggest that moving beyond chemical analysis was not so obvious as we may now presume.

Whatever the background, we know that Bernard's work

proved influential. No one claims that he invented experimentalism in physiology, but he gave a self-consciousness to experimentalists which proved powerful, not least in its extension to other biological sciences. Some botanists, especially in Germany, adopted similar methods for plant physiology. Famously (eggs again), some embryologists declared that the future of their science lay in experiments to control development, not just to elucidate the potentialities of the different 'elements' (germ-layers) of the embryo (Churchill, 1973; Gilbert, 1994). Jacques Loeb, who moved from Germany to the USA and there helped develop experimental studies on protozoa, was one of the theorists of experimentalism. As a positivist (and follower of the physicist Ernst Mach) he took an operationalist view of science – the ability to control was the measure of knowledge (Pauly, 1987). Indeed, from the late nineteenth century, so powerful was 'experiment' as a banner that much work was so labelled which might more accurately have been described as microscopical analysis.

That elision was evident in Britain and the USA from about 1870 as T. H. Huxley and his protégés campaigned to introduce 'biology' into universities. Huxley, 'Darwin's bulldog', was a classical zoologist and palaeontologist, known for his morphological studies, but he was also an ardent advocate of a wide-ranging science of form and function that was sometimes called 'experimental biology'. He arranged that schoolteachers should be trained to teach it, that medical students should learn and practise 'experimental physiology', and that the 'new botany' should be introduced into departments previously devoted to plant classification. Though the 'unification' of biology was rarely achieved in British universities, the movement for 'experiment' succeeded in as much as it drew on strong contemporary movements for the reform of medical education and for science in elementary education (Desmond, 1994; 1997; Geison, 1978; Gooday, 1991b; Thomason, 1987). From the end of the century it grew rapidly when government began to see 'science' as the key to the development of the Empire. Tropical medicine became a British specialism – parasitology, economic entomology, mycology, etc.,

took shape in this context and continued to be strongly supported, for 'home' as well as imperial reasons, through to the Second World War (Worboys, 1976; 1998).

These new biological sciences were experimental in a rather catholic sense. In the UK and the USA, proponents of the 'new botany' or 'new biology' or 'experimental zoology' often saw themselves in opposition to older naturalists – amateurs or museum curators whom they tended to dismiss as merely observational or taxonomic, mere 'stamp-collectors' (Maienschein, 1991). In fact, much of the new biology was microscopical observation or analysis, e.g. working out the life cycles of fungi – the stuff of classical zoology and botany in the German universities. Experiments in the Bernardian sense were a small part of the repertoire, but they were played up. If you stressed the manipulations involved in dissections or microscopical studies, these too could be encompassed as 'experimental'.

The development of laboratory biology was roughly contemporaneous with the founding of university laboratories for experimental physics and then engineering (Fox and Guagnini, 1998; 1999). In all cases, I suggest, these sites chiefly provided protected environments for the development of *analytical* skills (and methods); they were a way of bringing some of the world into the college, but that was easier if you referred to the spaces as 'laboratories' rather than workshops, and stressed 'experiment' rather than just analysis (Fox and Guagnini, 1999: ch 3). The laboratories were chiefly for teaching, and experimentation in a strict sense was a relatively small part of the work, but it was ideologically and prospectively important because it established university laboratories as creating novelties rather than simply analysing the world outside. We see here, again, the politics of ways of knowing and the power relations between them.

Experimentation in physical sciences

Historians who have studied physics have usually focused on physical theory and especially on generalisations such as the doc-

trine of energy, briefly discussed in Chapter 4. Latterly some have given detailed studies of experimentation, especially in the seventeenth and twentieth centuries (e.g. Collins, 1992; Galison, 1987; 1997; Gooding, 1985; Shapin, 1994; Shapin and Schaffer, 1985). Yet we lack a general history of 'experimental' physics, a term in common use in the later nineteenth century to contrast with mathematical physics. Historical studies of measurement cover part of this field, but in a sense they serve to highlight the key question for this chapter: what was distinctive about physical experimentation, *beyond measurement*? How did the methods and methodological statements of 'physicists' relate to those of contemporary chemists and biologists? Is it fruitful for us as historians to look for analogues of synthesis in chemistry or Bernardian experimentation in biomedicine?

That such questions seem scarcely to have been asked reflects two features of most modern history of science, against both of which this book is directed. First, the disciplinary divides among the historians who deal seriously with the intellectual content of the sciences and, secondly, the presumed primacy of physical sciences – usually justified in terms of mathematical sophistication, theoretical rigour and experimental accuracy. Philosophers of 'science' have usually read 'science' as meaning physics, and historians of science have usually seen chemistry and biology etc. as trailing along behind 'physics'. The classical post-Second World War history of science by Herbert Butterfield is exemplary here. It focuses on the scientific revolution of the seventeenth century which was mostly 'physics', the chemical revolution was delayed until the end of the eighteenth century, and biology brought up the rear with the Darwinian revolution of 1859 (Butterfield, 1957). Note, too, how the landmarks are changes of theory rather than of method.

This story of hierarchy and of philosophical and historical precedence has some merit; certainly its prevalence needs to be explained, and we need historical explorations of the extent to which this hierarchy was felt by different kinds of researchers at different periods. But we also need research from the opposite supposition – that the various sciences at any given period shared

questions of method and methodology; that in some ways they proceeded in parallel. This 'hypothesis of synchronicity' seems especially promising for the nineteenth century, after many new analytical disciplines had taken shape together. Whether we look to Paris as the world centre of the analytical sciences, or to London and the work of Humphry Davy and Michael Faraday at the Royal Institution, we have a sense of analytical researches developing in parallel across a broad front, with no clear differentiation between physics and chemistry. Many different elements, including light, electricity and heat were under investigation (Knight, 1986; 1992).

This specific feature of the time is clear in the contemporary discussions of the 'classification of the sciences' (Fisher, 1990) and in that formative text on 'scientific method', John Herschel's *Preliminary Discourse on the Study of Natural Philosophy*, first published in 1830. The son of William Herschel, the astronomer who classified stars, John was a major protagonist for 'science' and for 'scientists', as some of his friends had begun to call themselves. As an admirer of Francis Bacon he did much to create a self-conscious experimentalist tradition in England, but he was also very clear about the role of analysis in the constitution of the sciences around him: 'In pursuing the analysis of any phenomenon, the moment we find ourselves stopped by one of which we perceive no further analysis, and which therefore, we are forced to refer (at least provisionally) to the class of ultimate facts, and to regard as elementary, the study of that phenomenon and of its laws becomes a separate branch of science' (Herschel, 1835: 93).

From the middle of the century it is easier to talk of 'physics', when elements such as heat and motion were covered by more extensive notions such as energy, and as more interactions between elements were uncovered. In German universities, *Institutes* for physics accompanied or followed those for physiology; chemists usually led the way. In late nineteenth-century industrial Britain, the development of university departments usually followed this same order but, in Cambridge, chemistry followed the other two. In all these sites, 'experimental physics'

was among the new disciplines created by the nineteenth century, but it was usually experimental in the wide sense discussed above for biology – much of the work was analytical.

‘Natural philosophers’ were indeed much concerned with measurement of these elements such as light and heat, but some ‘experimental philosophers’ also focused on *interactions or reactions* between elements – the chemical elements or heat, light and, especially, the various kinds of electricity. Here, perhaps, one can see the beginnings of experimentalist traditions that were to produce ‘the new physics’ of radioactivity etc. at the end of the century. They were marked off from the Baconian experimentation of the eighteenth century by the focus on new ‘elements’; they were both analogous to and continuous with experimentation in chemistry (and engineering). For Britain, the founding father of this tradition was Michael Faraday, the patron saint of later experimentalists (Cantor, 1991; Gooding and James, 1985; Williams, 1971, 1987).

Faraday, the son of a blacksmith, became assistant to Humphry Davy at the Royal Institution (in some ways the London equivalent of the Collège de France). There he acquired the skills of a chemist, and followed up Davy’s work on electrochemistry. Faraday was no mathematician and he was not primarily interested in measurement. He did lots of analytical work, not least for the government, but his public work centred on searching out possible interactions between ‘elements’ (partly because of a metaphysical belief that they were but different expressions of a common force).

Historians of physics tend to stress this metaphysics and the way which he ‘intuitively anticipated’ field theory. Chemists have found Faraday’s style more natural, because of his cataloguing of experiments. Here it may be instructive to recall how Faraday systematically checked out the electrical properties of many materials (paramagnetic and diamagnetic, insulators and conductors). At one stage he worked through the various ‘kinds’ of electricity (electrostatic, voltaic, magnetic, thermal, fishy, etc.) to establish their identity. His most famous works concerned the interactions, assumed to be reversible, between the various

elementary forces – magnetism and current electricity, electrical forces and motion, electricity and polarised light, electricity and crystalline structure, light and electrochemistry, etc. He discovered many such interactions or ‘effects’, and he ‘prepared’ the discovery of others, accomplished later by researchers who employed more powerful equipment or happened to use the particular materials which ‘worked’. I suspect that much of the experimental physics of the nineteenth century could be expressed in a matrix of real and possible interactions between these ‘elements’ or ‘forces’. Among Faraday’s contemporaries and friends one sees a similar pattern in the work of Gustav Magnus, who in the laboratory at his home created the Berlin tradition of experimental physics (Fox and Guagnini, 1998; Kauffman, 1974).

Faraday’s successor at the Royal Institution was John Tyndall, a pugnacious ally of T. H. Huxley in their many struggles to secure more resources for scientists and less authority for bishops (Desmond, 1994; 1997; Turner, 1980). Tyndall, again, was an experimentalist; he had worked with Magnus in Berlin, and he ranged widely across the (analytical) disciplines. He followed up Faraday’s work on magnetic properties of various materials, involving the compression of hundreds of crystalline substances. These studies led to research on compression of ice (and glacier movement) and on the effects of solar radiation and heat radiation on atmospheric gases; hence to the scattering of the sun’s rays on dust, to the amount of dust in atmospheres, to floating organic matter and to bacteriology (Macleod, 1976). We note here the preoccupation with ‘outdoor physics’ or the physics of ‘nature’. We shall return to dust, as it were, and to nature, but first a glimpse of Paris and the French tradition.

In 1838 a Chair of Physics was created at the Museum of Natural History, which we have already visited for zoology, botany, geology, crystallography and chemistry. For the rest of the century, the chair of physics was held by three successive members of the Becquerel family. Antoine (1788–1878), for whom the chair was created, worked on diamagnetism before Faraday; he investigated the generation of electricity by pressure

on crystals, and by heating minerals; he also experimented on the synthesis of new crystalline forms of electric fields. From the 1870s the laboratory received state support for the development of electrical studies (Fox and Guagnini, 1998: 111–12). His son, Alexandre, became a world expert on the relationships between light and chemical reactions. His grandson, Henri, did further work on the effects of magnetism on polarised light, and on crystals which absorb or emit light; in 1896 he used X-rays on luminescent crystals – and thereby discovered radioactivity, the basis for an unexpected expansion of experimental physics in the twentieth century.²

We note here the continuity between this experimental physics and other studies at the museum, especially crystallography. We see how such studies of ‘effects’ could illuminate general theories (e.g. of the relationships between light and electricity), but that they were also investigations of natural curiosities (such as photo-luminescence), and the basis of useful instruments (such as the actinometer which used chemical reactions to measure quantities of light). We also see how ‘the new physics’ of radioactivity sprang from this long tradition of investigating ‘reactions between elements’ – experiments which ‘bridged’ chemistry and physics. We can see another aspect of that bridging if we examine the physics of clouds. Here I take my lead from an article by two American historians of modern physics, Peter Galison and Alexi Assmus (1989), but I develop the argument in terms of ‘ways of knowing’.

On clouds, dust and control

At the start of the nineteenth century there was much interest in the forms of clouds, not least among landscape artists such as John Constable. (And at the end of the century, John Ruskin continued the tradition.) We could call this ‘morphology’, in a loose sense, but it was not much more than a taxonomy of appearances, a matter of natural history. To *explain* the conditions under which clouds formed you needed to know about the

solubility of water in air at various pressures and temperatures, but this *analytical* understanding became so commonplace among chemists and other investigators that *we* hardly notice it. Increasing experience showed, however, that the simple analytical framework was insufficient – as gases were cooled or pressures lowered, water did not condense out unless particles were present around which droplets could form and produce ‘clouds’.³

Cloud formation was studied by John Aitken, a Scot trained in engineering, who devoted himself to experiments in natural philosophy – one of the many unpaid ‘devotees’, like Joule and Darwin, who produced much of Britain’s science in the nineteenth century. The study was continued by C. T. R. Wilson at the Cavendish Laboratory, founded in Cambridge in the 1870s as a home for measurement and for experiment (Schaffer, 1992; Sviedrys, 1970; 1976). Wilson came from the Lake District and was fascinated by meteorological phenomena, which he modelled in the laboratory. When he later discovered that ions could also act as centres of droplet formation, his ‘cloud chamber’ became an instrument for investigating charged particles – the key tool in the Cambridge investigations of particle physics. Wilson continued to focus on the *analytical* uses of the cloud chamber, while his more famous colleagues built experiments around it, and so helped build atomic physics. Galison (1997) has explored at length the continuing interactions and tensions between analysts, experimenters (and theorists) in twentieth-century physics.

But to grasp the shape of nineteenth-century physics we must not rush to the twentieth, rather we should ask about the ways of experiment and their relations to other ways of knowing and working. Here we note the recurrence of ‘dust’, already mentioned in connection with Louis Pasteur and John Tyndall. Aitken and Wilson also performed classic ‘dust’ experiments of the kind that Chevreul called *a posteriori* and Bernard called control experiments. All four, independently investigated ‘air’ from which dust had been removed, and then the effects of ‘adding dust’ – whether on cloud formation or on the fermenta-

tion of organic matter. These 'synthetic' experiments confirmed the role of 'dust' in these various phenomena – they confirmed the significance of a particular 'element' in the analytic understanding of these phenomena. The authors would have recognised their method in John Herschel's Baconian text, to which we referred above: 'We make an experiment of the crucial kind when we form combinations, and put into action causes from which some particular one shall be deliberately excluded, and some other purposely admitted; and by agreement or disagreement of the resulting phenomena with those of the class under examination, we decide our judgement' (Herschel, 1835: 151).

Adequate *analysis* of 'causes' allowed the widespread development of experimentalism. As Faraday noted: 'As an experimentalist I feel bound to let experiment guide me into any train of thought which it may justify; being satisfied that experiment, like analysis, must lead to strict truth if rightly interpreted; and believing also that it is in its nature far more suggestive of new trains of thought and new conditions of natural power' (Thompson, 1898: 242).

But we note two further characteristics of these experiments on clouds that bring them alongside the chemical syntheses and the biological experiments in a way that seems stranger for physics because of the usual emphasis on abstraction, theory and mathematics. The cloud experiments could be models of real clouds; they were 'mimetic', to use Galison's term. (Would one need a term for chemical or biological experiments where mimesis seems more natural?) One way of using a cloud chamber was to play with it to reproduce natural phenomena – that aspect fascinated Aitken, and Wilson never lost his interest in meteorology. Like chemists 're-creating' an organic compound, or physiologists inducing diabetes in an experimental animal, physicists were modelling 'nature' – with the potential for control.

This kind of experimentalism was probably more widespread than most histories of physics suggest. If we look for case histories in Manchester c.1880, in the first of the English provincial universities, we find the academic engineer, Osborne Reynolds,

very much interested in 'outdoor physics'. He is known for his analyses of flow and lubrication (Reynolds's numbers), but he also modelled river estuaries, and (for fun?) he showed how it was that lightning could explode trees (by boiling the sap). His chemist colleague, Henry Roscoe, and the physicist, Arthur Schuster, were experts on spectroscopy, reproducing in laboratories spectroscopic patterns resembling those observed from heavenly bodies (Schaffer, 1995). The techniques used in the analysis of natural or technological phenomena, could also be used in experimental re-creations, but the experiments could also become analytical techniques – in some contexts, for some purposes, experiments could be subordinated to analysis (Davies, 1983; Kargon, 1977).

Here we recall the remarks in my first chapter about the nesting of 'ways of knowing', and the different possible hierarchies. We are now in a position to expand that discussion and to reflect on the (contested) divisions of the sciences, on differences in status between such branches, and on the ways that such status differentials have varied over time and between sites. These reflections will also serve to introduce the final section of the chapter – on the relations between experiment and invention.

Experimentalism and hierarchies of knowledge

We noted that Alexandre Becquerel, by studying the effects of light on chemical reactions, created an actinometer that could serve to measure light. Tyndall's work on dust in air, and the work on cloud formation, gave rise to instruments for measuring the quantity of dust in air (and thus to the investigation of fogs, smogs and airborne infections – major problems for Victorian cities). Experiment-as-analysis is a key part of the analytic-synthetic dialectic, and of the relation between laboratory studies and investigations of 'the real world'. These problematic relations were formulated in different ways in different disciplines, reflecting the different power relations.

Sometimes studies of 'outdoor physics' or of 'technological

systems' (and the associated instrumentation) have been known as 'applied physics' (and downgraded thereby), but perhaps we would do better to use another terminological frame, common in medical sciences and chemistry – the contrast of 'general' and 'special', e.g. general pathology and special pathology, where 'general' covers such processes as inflammation, and 'special' refers to particular diseases or the lesions of particular organs. Chemists also referred to general chemistry, e.g. the study of reaction mechanisms, and to special chemistry, e.g. of chlorine compounds. In these formulations the contrast is not between pure and applied, but between a push to general explanation and a concern with explaining the complexities of the world; between, in my terms, the study of the properties of 'elements' and the properties of particular 'compounds or systems'. We do well to remember how much 'physical science' could have been called 'special physics' because it was about 'compounds'. And so it remains.

Of course, the distinction between elements and compounds is relative. Part of the progressiveness of science is the potential to envisage 'elements' as 'compounds' (or as derived from a fundamental element), for example to see atoms as 'planetary systems' of 'subatomic particles'. The 'higher destiny' of the cloud chamber was to be a detector for charged particles, a means of analysing the disintegration of atoms (and a means of establishing that disintegrating atoms also existed 'in nature'). So analysis and synthesis could be reconstructed in dialectical relation at a deeper level, and the 'general' to 'special' fulcrum shifted accordingly.

So consider the remark of Rutherford, Schuster's successor as Professor of Physics in Manchester – that science was either physics or stamp-collecting. He was claiming hegemony not only for experimentalism over (mere) analysis and natural history, but also for the pursuit of the general over that of the 'special', and for his new physics over the electrotechnics which occupied much of 'his' building in Manchester. As we shall see in the next chapter, Schuster had regarded the electrotechnics as vital and had argued for continuities between workshops and university

laboratories, but Rutherford was not interested; the electrical work could be done in departments of *electrical engineering* in universities or in technical colleges (as was indeed happening in Manchester) (Wilson, 1983). For him, the constituents of atoms were the centre of 'science', an attitude which remained common for most of the twentieth century. It was reinforced by the link between atomic physics and astronomy, and thus with explanations of the infinitely vast, powerful and primordial. After the Second World War, the atomic physics which had once served as paradigmatically 'useless', promised cheap, clean energy and military dominance – as well as being fundamental. This view of fundamentals, when sold to governments by a generation of physicists bound together by their war work, created huge international facilities for experimental physics, e.g. CERN at Geneva. Now, however, as this vision has come to be rivalled by visions more biological and commercial, we can see more clearly how it was constructed, and how and where experimentalism was contested.

For example, as we have seen, one of the key sites for experiment was physiological laboratories of the later nineteenth century, where chemical and histological analysis was mostly subordinate to the experiments on live animals conducted by the head of the physiology institute. But suppose the head of the institute was a chemist or a histologist, rather than an exponent of vivisection. Under such circumstances the focus might be analytical and possibly experiments might be undertaken to produce new substances for analysis, e.g. to complete an analytical series. Or, when biological substances must be measured (and standardised) in the absence of chemical tests, researchers resort to routine 'experiments' called bioassays – say to measure the effects of 'nerve-growth factor' on the spinal ganglia taken from early embryo chickens. In such bioassays, the aim is to measure the substance, not to explore the effect; 'experiment' here serves analysis.

More generally, in clinical contexts such as hospital laboratories, the analytical values of diagnosis and therapy may be predominant and experimentalism may be subordinated. In early

nineteenth-century France, the Paris Academy of Medicine promoted physiological researches on the mechanisms of vomiting in animals. In a university laboratory these might have been a means of exploring reflexes, for clinicians they were orientated towards elucidating the conditions that provoked vomiting in patients (Lesch, 1988). Similarly, electrical stimulation of nerve and muscle could be a means of exploring normal and abnormal reaction mechanisms, or a means of clarifying neurological and muscular disorders. Clearly these orientations are matters of degree, but subtle differences of this kind are often important in the politics of knowledge and in determining the development of research in particular sites. The history of 'clinical research' or 'research in engineering' suggests that in institutions oriented to practice, analysis has often been the preferred mode of science, and in such situations 'experimentalism' may be subordinated.

The details do not matter much, but the general point is important. In discussing scientific projects we can analyse the component ways of knowing and the logical relations between them, but we also need to assess the 'direction' of the project – its leadership and its orientation in terms of the predominant way of knowing. If we seek an explanation for the overall form, then we must look to the contingencies of social life and to the politics of knowledge as favouring certain modes over others. Much the same applies to the relations of experiment and invention.

Experiment and invention

In preparing to define experimentalism in my specific sense of 'synthesis', I also related experiment to natural history (experimental histories) and to analysis (analytical measurements, etc.). Similarly, when we try to pin down the specificities of invention as a major function of nineteenth-century technology, we need to consider a comparably wide range of meanings, where invention appears alongside the other ways of making (and knowing) – alongside craft (and natural history) and rationalised production

(and analysis). I want to suggest that crafts are generally inherited rather than novel, and that rationalised production in some sense reconstitutes existing processes, but that (synthetic) invention like (synthetic) experimentalism creates novelties – things and processes that did not exist before. That is the core of my usage, though clearly, definitions are slippery.

Some ‘inventions’ may be regarded as ‘findings’ – from ‘*invenire*’, ‘to light upon’, a usage in early modern English.⁴ Roast pork may have been invented (or discovered) in a burnt pig-sty, some medicines were discovered by chance. By ‘torturing’ materials, in Francis Bacon’s sense, we may find new and useful properties, e.g. new kinds of hardened pots or better curing of leather. Here we see the links with natural history and craft, and in as much as more sophisticated inventions comprised many minor adjustments attained by trial and error, so this simplest kind of invention may indeed be fundamental (Basalla, 1988). But usually invention has involved some sort of *analysis*, some attempt to assess a ‘need’ or an economic possibility. Why, then, do I *contrast* invention with analysis and rationalised production? Was not James Hargreaves’s spinning jenny an ‘invention’ (see O’Brien, Griffiths and Hunt, 1996)?

The spinning jenny modelled the actions of spinning on a wheel, it mechanised the twisting and the backward-and-forward motions, and allowed one person to work several threads simultaneously. One suspects that many people had thought of ‘multiplying’ spinning in this way, but doing so successfully involved craft skills and persistence. We could call it an *analytical invention*; the industrial world is full of them, because so many of its commodities are mechanically produced substitutes for prior craft products. Of course, you can argue as to when analytical inventions move beyond ‘reconstruction’ or multiple replication and so produce ‘novelty’. The mills of Manchester in 1800, seven-storey palaces packed with machines, were novel in many senses; they involved much ingenuity, and called for management (another novelty); but still they mostly reconstituted craft processes.

Yet clearly, in some contexts, inventions can be *synthetic* nov-

elties, intended to meet aims imagined but never realised. Many such goals, e.g. human flight, persisted for centuries before they were attained. Like synthetic experiments, invention may be led by hypotheses about new worlds, or worlds where old ends required new means.

'Invention is a new combination from "the prior art"' says a curious book on the sociology of invention, full of aphorisms, most of which seem rather obvious (Gilfillan, 1962). But they will serve here to stress the idea of combinations or synthesis, at the level of materials or processes, resources and purposes, as in the curious case of barbed wire. It was invented in the great plains of North America as they were developed for cattle ranches; fences were needed to restrain the movement of cattle, but wood and stone were rare and the lengths of fence required were great. Two methods were tried: one was traditional in other places – thorn hedges; the other was an adaptation of a new industrial product – (plain) wire made into fences. Neither worked well, but then someone thought of the hybrid. Wire fences with wire 'thorns' worked (and sold) very well indeed (Basalla, 1988: ch. 2).

Such examples show a strong analogy between synthetic invention and synthetic experiment, but obviously there are many differences other than of overall purpose. The components of barbed wire were everyday objects, not the objects of detailed analysis or any other form of close study; it seems natural to describe it as an invention, and its initial use as a 'trial' rather than an experiment. But of course, the relations may change over time. Inventions, like any other technological artefacts, may prompt analysis for the rationalisation of production and use, or indeed raise problems that attract investigators who are building systematic knowledges. No doubt the compositions and dimensions of barbed wires have now been closely analysed to effect the greatest economy in meeting any particular use, and perhaps the form of the barbs has been refined from studies of the distribution of nerve endings in bovine skin. (Who knows – perhaps there are technoscientific networks for 'bovine-kinetics'?)

But if synthetic invention may prompt analytical (or experimental) researches, what shall we say of the converse relation? What can we say about analytical studies as (re)sources for synthetic invention?

Perhaps we have already given the core of an answer in our discussion of analysis and experimentalism. The elements discovered by analytical sciences were *generalisable* – that is what made them ‘elements’ for science, not just ‘elements’ of a particular craft process such as weaving. So these elements could be put together in new ways – not just as synthetic experiments (for light), but also as ‘synthetic’ inventions (for fruit). The massive extension of analytical STM from about 1800 made available many new procedures for handling ‘elements’, which were of use to inventors; and the experimentalism we have discussed, from Faraday through to late nineteenth-century universities, produced ‘effects’ and equipment, such as model electric motors, which inventors could play with and develop for economic uses. Many of the classic electrical inventions of the later nineteenth century were derived from laboratory devices – put together in various ways to produce ‘economic’ effects. In such cases, we begin to see, not only the similarities between synthetic experiments and inventions, but their reciprocal, creative and generative relations. Model systems that could produce both light and fruit formed the core of new (synthetic) technologies.

These later nineteenth-century symbioses between science and technology did not mean that all inventors were graduate scientists, but they suggest why up-to-date intelligence from university laboratories eventually became a desirable commodity, and why some academics also benefited from the interaction. Collaboration between universities and industry was formative for a few areas of STM at the end of the nineteenth century, notably in electrotechnics and synthetic dyestuffs-cum-pharmaceuticals; in the twentieth century it became crucial to universities, companies and governments. The growth and varieties of these interactions, but also their limits, are the subject of the next chapter.

Notes

- 1 On Pasteur, see Geison (1970–80; 1995), Latour (1987a) and Saloman-Bayet (1986).
- 2 For the various Becquerels see the *Dictionary of Scientific Biography*.
- 3 The conditions of formation of dew, as investigated by Dr Wells, was one of John Herschel's key examples of scientific investigation. See Herschel (1835).
- 4 For stimulation on the relations of experiment and invention I am indebted to a (then?) unpublished paper by Johann Martin Stoyva.